LEAD SCORING CASE STUDY

Turning 9,240 Website Leads Into a Prioritised Sales Pipeline

A logistic regression model for X Education, scoring every lead 0–100 so the sales team calls the right people first.

Prepared by **Vishnuvardhan Reddy Janagam**

IIIT Bangalore | Data Science with Generative AI | Logistic Regression Assignment

38.5%
Base conversion rate

0.976
Test-set AUC

91%
Conversion among hot leads

THE BUSINESS PROBLEM

X Education Converts Only 1 in 3 Leads It Chases

Every lead — hot or cold — gets the same amount of sales effort. The CEO's target: focus calls on the leads most likely to buy, and lift conversion among those calls to ~80%.



Only 38.5% of leads convert — the rest absorb calls the sales team can't get back.

WHAT THE MODEL NEEDS TO DO

1. **Score every lead 0–100**

   so a higher score means a higher chance of buying

2. **Explain the score**

   logistic regression, so the sales team can act on it

3. **Adapt to changing goals**

   aggressive calling in intern season vs. minimal calls once targets are met

APPROACH

Seven Steps From Raw Leads to a Ranked Call List

01

Clean the data

Handle 'Select' placeholders, missing values, outliers, rare levels

02

Explore (EDA)

Find which variables actually separate buyers from non-buyers

03

Prepare for modelling

Dummy variables, 70:30 split, scale numeric columns

04

Build the model

RFE (CV-tuned) → drop by p-value → drop by VIF

05

Evaluate

ROC/AUC, pick a cut-off, confirm on the untouched test set

06

Score every lead

Probability × 100 = Lead Score; hot leads above cut-off

07

Answer the business questions

Move the cut-off for intern season / target-met scenarios

Assumption: Tags are set by sales during follow-up, so the main model is a nurturing-stage scorer — a no-Tags model is also tested for brand-new leads.

DATA PREPARATION

Cleaning Turned 37 Raw Columns Into 15 Reliable Ones

“Select” = missing

Dropdown placeholders (e.g. Specialization, City) carry no information — converted to null before anything else

Outliers capped

TotalVisits and Page Views per Visit winsorised at the 99th percentile

Dropped 21 columns

7 columns >40% missing (Lead Quality, Asymmetrique scores…) and 14 with one level covering >95% of rows

Rare levels grouped

Any category level under 1% of leads folded into ‘Others’, so dummy columns stay stable

Missing ≠ random

Blank occupation converts at 14% vs 49% filled — so gaps are kept as their own ‘Not Specified’ level, not imputed away

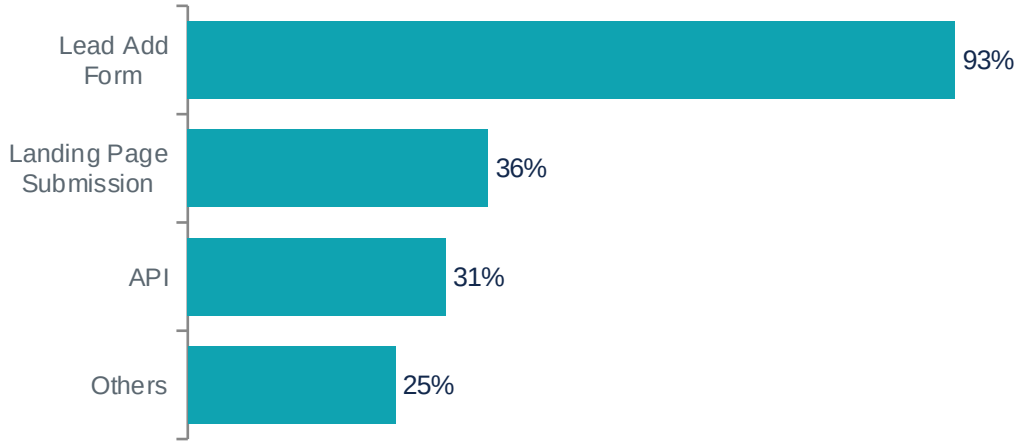
One new metric

Avg. Time per Visit = Total Time ÷ Visits — a cleaner read on engagement per session

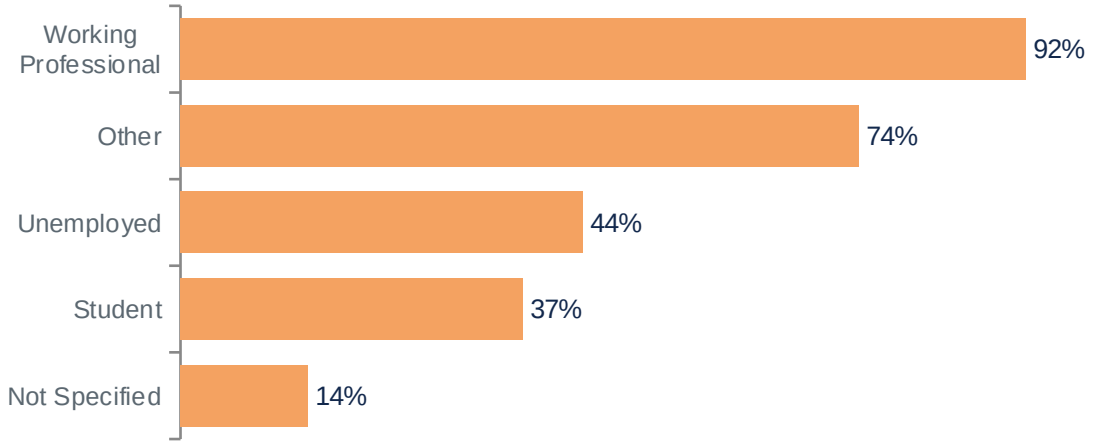
EXPLORATORY DATA ANALYSIS

Origin, Occupation and Engagement Separate Buyers Early

Conversion rate by Lead Origin



Conversion rate by current occupation



ENGAGEMENT: TIME ON THE WEBSITE IS THE STRONGEST NUMERIC SIGNAL

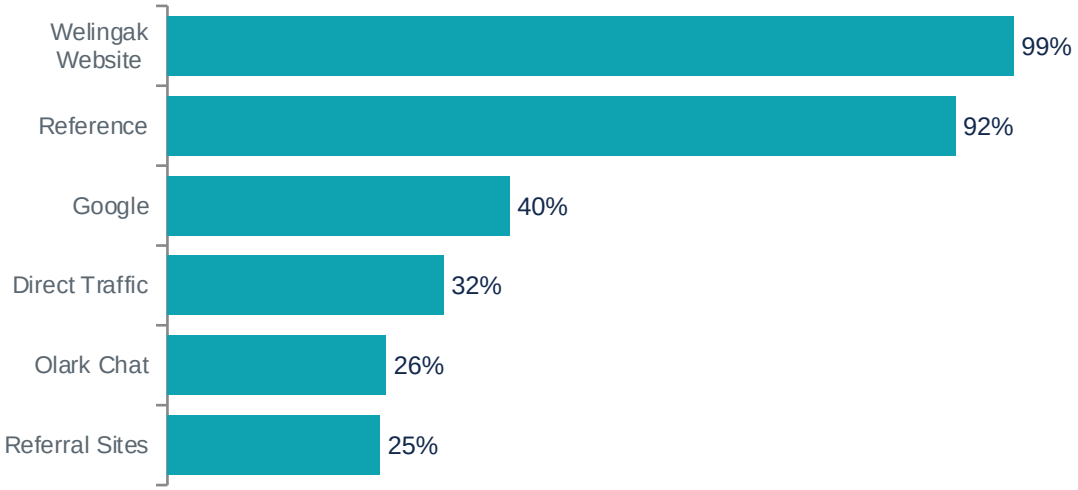
739 seconds average time on site for converted leads, vs **330** for non-converted (correlation 0.36 with conversion — the highest of any numeric variable).

Sales Tags and Lead Source Show the Widest Spread

Conversion rate by sales Tag (top / bottom 5)



Conversion rate by Lead Source (selected)



Tags are recorded by sales during follow-up, so they confirm an outcome already in motion — treated as the strongest signal, not a lever, in the recommendations ahead.

MODEL BUILDING

From 70 Candidate Variables to a Lean, Significant 23

70 variables
after dummy encoding

→

RFE (n=25)
tuned by 5-fold CV
AUC-maximising

→

Drop by p-value
remove weakest
until all p < 0.05

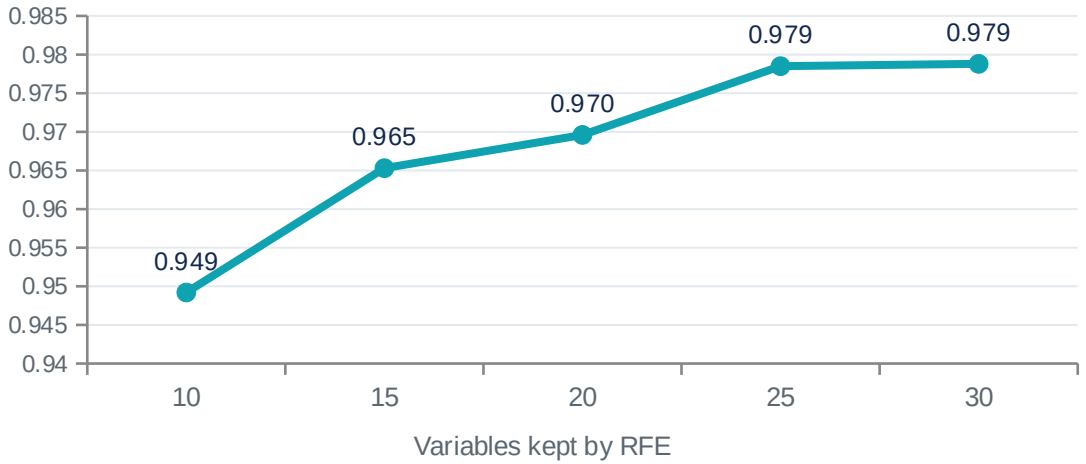
→

Drop by VIF
remove most collinear
until all VIF < 5

→

23 variables
final model

Cross-validated AUC by number of RFE features (train set)



WHAT SURVIVED THE FINAL CUT

- Do Not Email, Total Time Spent on Website
- Lead Origin (Lead Add Form) · Lead Source (Olark Chat, Welingak Website)
- Occupation (Student, Unemployed, Working Professional)
- Last Activity (SMS Sent) · Last Notable Activity (Email Opened, SMS Sent, Others)
- 11 Tag levels (sales follow-up status)

Every remaining variable: p-value < 0.05, VIF < 5 — no redundant or insignificant predictors left.

MODEL EVALUATION

0.976 AUC on Unseen Data — and No Sign of Overfitting

ROC curve — test set



AUC	Accuracy	Sensitivity	Precision
0.976	**93.5%**	**92.9%**	**90.5%**
Test	Test	Test	Test
0.980 train	93.9% train	93.3% train	91.2% train

TRAIN VS TEST GAP < 1 POINT ON EVERY METRIC

The model generalises well to leads it never saw during training — confidence it will hold up in production.

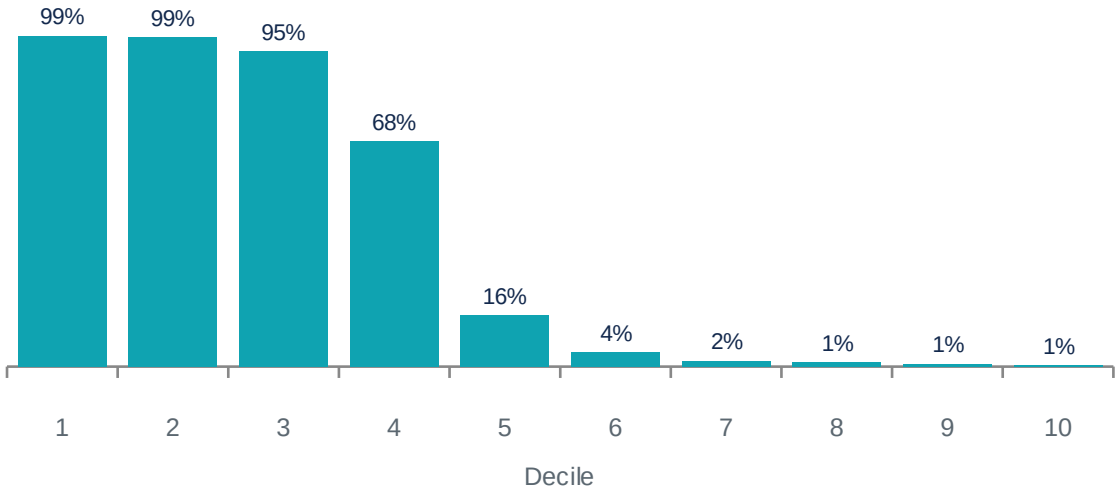
Comparison with other models (test set)

Model	AUC	Accuracy	Sensitivity
Final logistic regression (23 vars)	**0.976**	**93.5%**	**92.9%**
Logistic regression — all 70 vars	0.976	93.7%	92.9%
Random forest (tuned)	0.971	91.1%	92.8%
Decision tree (tuned)	0.965	92.2%	89.5%
Logistic regression — without Tags	0.884	80.5%	82.5%

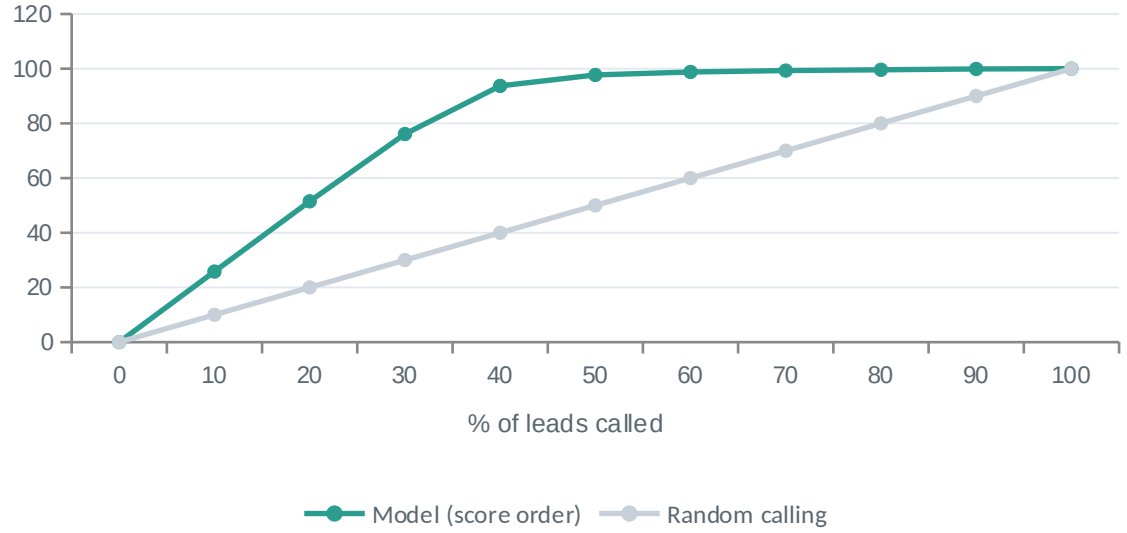
LEAD SCORING

The Top 30% of Leads Hold 76% of All Conversions

Conversion rate by score decile (1 = highest scores)



Cumulative gains — model vs. random calling



3,633

hot leads (score > 35)
39% of all leads

91.2%

conversion rate among
hot leads (target: 80%)

2.5x

lift vs. random calling
in the top 3 deciles

RECOMMENDATIONS · WHAT DRIVES CONVERSION

Three Levers the Business Can Pull Right Now

24x

Grow the Welingak Website channel

Leads from this partner convert at 98.6% vs 40% for Google — odds x24. Prioritise it in marketing spend and give its leads first call.

14x

Target working professionals

91.6% conversion vs 43.6% for the unemployed — odds x14. Tailor messaging and ad targeting toward this segment.

5x

Push visitors to the Lead Add Form

92.5% conversion vs 36% for landing-page submissions — odds x5. Make the form more prominent on the site.

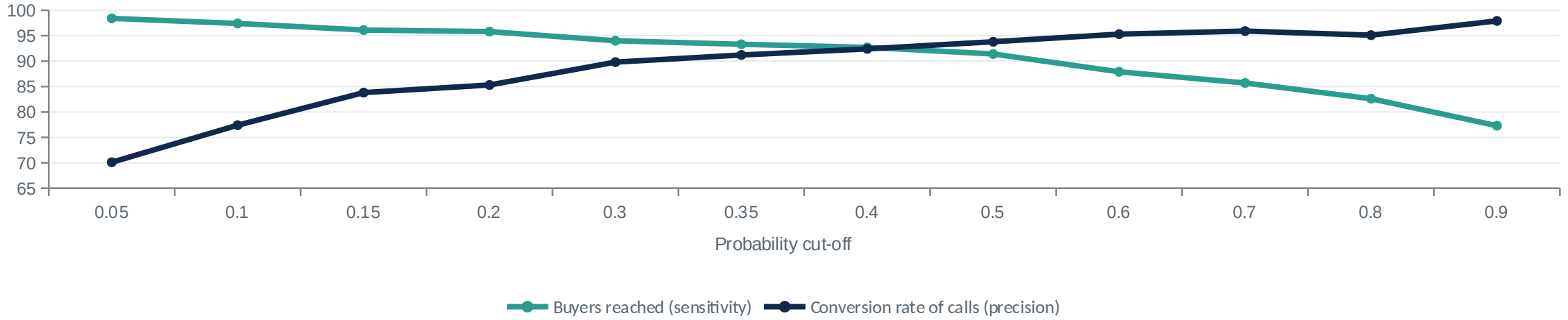
ALSO WORTH NOTING

Last Activity = SMS Sent (odds x4.5) is fully within the sales team's control — a next-best lever once the top three are in motion. Make the occupation field mandatory: a blank answer alone flags a 14% conversion segment.

RECOMMENDATIONS · FLEXING THE MODEL

One Lever — the Cut-off — Adapts to Either Scenario

Moving the probability cut-off changes precision vs. reach (test set)



cut-off 0.15

Q3 · Intern season

- Calls: 1,219 (+123 vs normal)
- Buyers reached: 96.1%
- Conversion of calls: 84.2%

cut-off 0.35

Normal operation

- Calls: 1,096
- Buyers reached: 92.9%
- Conversion of calls: 90.5%

cut-off 0.80

Q4 · Target already met

- Calls: 922 (−33% vs normal)
- Useless calls cut 57%
- Conversion of calls: 95.1%

CONCLUSION & NEXT STEPS

A Model the Sales Team Can Trust — and Steer

- **93.5% accuracy, 0.976 AUC**

  on unseen leads, with less than a 1-point gap to training — no overfitting

- **91% conversion among hot leads**

  beating the CEO's 80% target, using just 39% of the calling effort

- **One dial for two scenarios**

  lower the cut-off for aggressive intern-season calling, raise it once targets are met

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NEXT STEPS

- Monitor monthly; retrain if hot-lead conversion drifts below 80%
- Standardise and speed up sales tagging — it is the model's main fuel
- Pilot the no-Tags model (AUC 0.88) for scoring brand-new leads before first contact